Finite marked subgroups by successive stabilizers

Human Compression project

Let G be a specified finite group, let J be its nonidentity involutions, and let A act on G by automorphisms. The action need not be faithful. A string is an ordered tuple $t = (t_1, \dots, t_m)$ with $[t_i, t_j] = 1$ whenever $|i - j| > 1$. Write $H(t) = \langle t \rangle$. Its intersection property is

$$\langle t_i : i \in I \rangle \cap \langle t_i : i \in K \rangle = \langle t_i : i \in I \cap K \rangle \quad (I, K \subseteq \{1, \dots, m\}).$$

A string satisfying this property is a C-marking of its generated subgroup. It need not generate the ambient G.

Lemma 1 (An exact extension test)**.** *Suppose t has the intersection property, $x \in J \setminus H(t)$, and x commutes with $t_1, \dots, t_{m-1}$. The extended string $(t_1, \dots, t_m, x)$ has the intersection property if and only if its suffix $(t_2, \dots, t_m, x)$ has it and*

$$H(t) \cap \langle t_2, \dots, t_m, x \rangle = \langle t_2, \dots, t_m \rangle.$$

This gives a recursive exact test, ending at length at most two.

Proof. We retain the standard end-intersection criterion for string markings: two end deletions with the intersection property, and their correct common-middle intersection, imply the property for the full string (McMullen–Schulte, Proposition 2E16). The first end deletion here is the assumed old marking. In the recursive suffix test its old prefix is a submarking of t, so again has the property. Two distinct nonidentity involutions have order-two cyclic subgroups with trivial intersection, giving the base case. The condition $x \notin H(t)$ ensures distinctness in every suffix. Induction now proves sufficiency. Necessity follows by applying the full intersection property to the displayed subsets and to every suffix. □

Let A_t be the subgroup of A fixing every entry of t, and put

$$P(t) = J \cap \bigcap_{i=1}^{m-1} C_G(t_i).$$

The empty intersection is all of J. Starting at the empty tuple, make the following recursion through a specified depth d. At t, form $P(t) \setminus H(t)$, take one representative from each A_t-orbit, and append each representative which passes Lemma 1. Update $H(t)$ and the pointwise stabilizer. Record a visited tuple whenever it has the desired invariant property, and continue until depth d or $H(t) = G$.

Theorem 2. *Before any optional pruning, this recursion visits precisely one representative of every A-orbit of C-markings in G of length at most d. Thus it classifies any A-invariant selected family among these markings. No extension of a tuple generating G is lost by stopping there.*

Proof. The candidate set is invariant under A_t, since its conditions are generation, membership and commutation with entries fixed by A_t. Every C-marking is independent: if a mark lay in the subgroup on the other marks, intersecting that subgroup with the cyclic subgroup of the mark would contradict the property. In particular each final mark lies outside its previous prefix subgroup and every prefix is a C-marking.

Induct on length. A marking of length $m+1$ has a prefix equivalent under A to one already visited. After making that change, its last entry is a candidate passing the exact extension test. An element of the prefix stabilizer sends it to the chosen orbit representative, so its full orbit is visited. Conversely, an automorphism between two visited extensions first identifies their prefixes. The induction hypothesis makes those prefixes equal; the automorphism then belongs to A_t, so their last entries have the same candidate orbit. This proves uniqueness. Independence forbids any extension once the prefix generates G. The fixed depth and finite candidate sets prove termination. □

Lemma 3 (A necessary completion test)**.** *Every extension of t generates a subgroup of $K(t) = \langle H(t), P(t)\rangle$. If a property required of a recorded marking cannot hold in any subgroup of $K(t)$, its whole branch may be discarded. In particular, when G acts on Ω, an intransitive $K(t)$ has no transitive completion. A proper $K(t) < G$ has no completion generating G.*

Proof. Every future entry has index at least $m+1$, hence commutes with the first $m-1$ old marks and belongs to $P(t)$. All completed generators therefore lie in $K(t)$. A subgroup of an intransitive group is intransitive, and a subgroup of a proper group is proper. The conclusion follows. After appending x, the new pool is $P(t) \cap C_G(t_m)$, independently of x. Caching these complete commuting sets therefore preserves the recursion exactly. The subgroup test must use the entire pool, not merely orbit representatives. Pruning changes the counts of visited prefixes; these are not then a census of all partial C-markings. □

Permutation application. Take $G = A_n$ and $A = S_n$, acting by simultaneous conjugation. This action classifies the actual permutation markings up to renumbering the points. Its stabilizer at t is the successive centralizer $C_{S_n}(t_1, \ldots, t_m)$. All even involutions are included: in the implementation they are the disjoint union of all order-two conjugacy classes of A_n. Every transitive C-marking under consideration consequently occurs, with no classification of transitive subgroups assumed. If a visited proper subgroup is already transitive, it is recorded and still extended. Transitivity is not a stopping condition.

Permutation equivalence differs from abstract marked-group isomorphism. The latter is tested by the homomorphism sending the ordered generators of one generated group to those of the other, with an explicit bijectivity test. Reversing the tuple is duality. An abstractly self-dual marking can have a permutation representation inequivalent to its reverse.

Attribution and computational premises. Enumeration by prefix stabilizers is established methodology: Leemans–Mixer, *Algorithms for classifying regular polytopes with a fixed automorphism group*, Contributions to Discrete Mathematics 7(2) (2012), Section 5, attributes it to Hartley–Hulpke (2010). The present proof specifies an ambient subgroup search, the recursive intersection test and the necessary completion test used by the accompanying program. No priority for the general orbit method is asserted.

The complete GAP program and exact execution are required finite premises for each application, including its ambient model, all involutions, candidate orbits and final tuples. Exact group generation, conjugacy classes, centralizers, subgroup intersection, orbits and homomorphism checks retain GAP's software correctness as an explicit premise. The polytope correspondence and the end-intersection criterion remain mathematical inputs. A saved count without a complete execution is not a census certificate; no Lean verification is claimed here.